Feng Wang

fengwang@stu.pku.edu.cn · 15112645887 · wfswechatid · caseclose

Education

Peking University 2024.09 – 2027.06

M.S. student School of Electronic and Computer Engineering Computer Science and Technology
Research interests: multimodal agents, reinforcement learning, multimodal understanding, code-driven data synthesis

Harbin Institute of Technology 2020.09 – 2024.06

B.S. Faculty of Computing / Honors School Information Security *Major rank*: **2/82** IELTS: 6.5 CET-6: 530

Internship

Tencent – WeChat (WXG) Technical Architecture *Research Intern* 2025.11 – 2026.08

1. Cognitive-structured Multimodal Agent (CMA) Paper Project Page Code

> *deployed to the WeChat Xiaowei Agent pipeline, currently in gray-scale testing*

- **Problem reformulation:** Multi-turn multimodal dialogue suffers from linear visual-token growth, interference from irrelevant images and unstable long-history reasoning. Observing that **textual history is continuous while visual history is sparse** — only a small subset of past images matters per turn — I recast the task as **visual-context selection**: learn a policy that picks a minimal relevant image subset, replacing full visual-context replay with retrieval-based access, via PAE (perceptual compression into episodic visual memory), CoRE (turn-level retrieval) and MEC (tool and multimodal-execution orchestration).
- **Data & training:** Built the Unified Scenario Engine to synthesize 20-turn dialogues with fine-grained retrieval annotations, bypassing manual annotation of long-horizon retrieval supervision; devised a CoRE SFT→CoRE RL→PAE RL pipeline (8-node distributed), where CoRE is rewarded by difficulty-weighted Jaccard and PAE is trained against downstream retrieval accuracy under a frozen CoRE, aligning the compressor with the task it serves rather than a caption-quality proxy.
- **Results:** Built M2CA-Bench (100 sessions / 2,000 annotated turns, bilingual, stratified along four difficulty dimensions, strictly held out). **PAE+CoRE lifts a 32B baseline from 81.9% to 92.0% (Full) and 62.0% to 83.5% (Hard) while cutting per-turn latency 23.1 s→12.7 s and raising generation quality 7.93→8.77**; an 8B model reaches 91.4% (Full) / 82.0% (Hard), and 89.6% on Chinese Full.
- **Ablations & comparisons:** The three training stages contribute 63.6%→70.9%→77.4%→82.0% on Hard; **against MemoryLLM- and ReasoningBank-style text-only memory, which reach only 24.4% / 39.0% on Hard, this design reaches 82.0%**, confirming that thumbnails must be retained alongside textual metadata; versus the concurrent agent framework GenArtist, generation quality decays by just 0.45 (9.51→9.06) over the second half of long sessions against its 1.97.
- **Engineering & C-side loop:** Delivered CMA-Harness (17 tools, multi-scope persistent memory, OpenAI-compatible APIs) with a read/write-split concurrency model — read-only tools fan out asynchronously while mutating tools run serially — and a three-tier retrieval cascade (zero-LLM heuristics → text prefilter → multimodal retrieval) to contain cost. **The architecture has been migrated to the WeChat Xiaowei Agent pipeline and entered gray-scale testing**, with **ongoing badcase attribution and iteration driven by real user-feedback data**.

2. InfoCodeGen: code-driven infographic generation

> *for the WeGen Omni V4 foundation model — CoT writes code, renderer produces the image*

- **Inference pipeline:** Built a generation path where the model **emits HTML/SVG/restricted LaTeX in the CoT and a renderer produces the infographic** — layout and copy are planned in the reasoning trace, then written as executable source and deterministically typeset, replacing pixel-level drawing with "reason, write code, render".
- **Problem & insight:** Infographic text must jointly satisfy correct glyphs, sound layout, stylistic diversity and controllable density; pixel-space methods (GlyphControl / TextDiffuser / AnyText) struggle under long

text and complex layouts and rely on OCR re-annotation. Proposed a **paradigm inversion: generate the source code that renders text, not the text pixels**, so training and inference share the same intermediate representation.

- **Data synthesis:** The same source is rendered to PNG via Playwright; **bounding boxes are read from the DOM via Range.getClientRects(), eliminating OCR re-annotation**. Samples are seed-replayable via style_spec; controlled source edits yield before/after pairs with $<$**0.1% pixel change** outside the target. Style is factored into **9 orthogonal control axes** (37 layout skeletons $\times$ 5 canvases $\times$ 4 density levels), with language-gated fonts and three quality gates: DOM checks $\rightarrow$ VLM visual review $\rightarrow$ pixel consistency.
- **Output:** bbox-annotated samples and doubly-verified editing triplets spanning 6 languages, 5 canvases and 4 density levels, with **OCR-free annotation and low-cost scaling**, **feeding infographic training for WeGen Omni V4** and the inference pipeline above.

3. WeGen Omni V3 unified understanding-and-generation foundation model

> *domain data & benchmark construction (scientific figures / infographics / e-commerce); deployed to Weishi Chuan and Tencent Games; open-source and tech report in early September*

- **Data production:** Led pre-training and SFT data development for this foundation model across scientific figures, infographics and e-commerce images via an end-to-end pipeline of rule-based filtering, VLM screening, OCR/caption consistency checks, quality evaluation and iterative refinement; **distilled 32M+ raw images into** $\sim$**4.9M high-quality training samples**.
- **Evaluation system:** Built a text-rendering benchmark (1,000 ground-truth items across 10 scene types $\times$ 5 text lengths $\times$ 4 difficulty levels) scored by **two independent read-back paths — OCR and multimodal — for cross-validation**: the former separates over- from under-writing, the latter adds legibility and reading-order diagnostics. **Deliberately refused to collapse them into one weighted score**, so the loosest metric cannot decide a batch; slice-level attribution then feeds back into data composition and caption rules.
- **E-commerce track:** Rather than sampling at the natural frequency of collected data, **anchored coverage to the JD/Taobao market-wide SKU distribution** to balance the long tail, distinguishing “missing volume” from “missing hard cases”; to enforce the hard constraint that product identity must never be altered, built a consistency checker scoring **ΔE (CIEDE2000) color drift, logo detection with feature matching, OCR edit distance, and related criteria** per dimension. Further built **530K editing instructions across 14 task types** supporting SFT on 5.56M samples.
- **Deployment & open source:** Model capabilities have been applied to WeChat’s e-commerce virtual try-on mini-program **Weishi Chuan (微试穿)** and Tencent Games; **the model will be open-sourced with a technical report in early September**.

Baidu – VIS, Multimodal Perception & Understanding *Research Intern* 2025.04 – 2025.09

- **Problem & insight:** GUI agents must emit both an action and a coordinate, yet current models fail in two ways — misreading intent under ambiguous high-level instructions, and drifting off target on high-resolution, densely packed interfaces. Proposed **H2L-G1**, a tool-augmented reasoning framework that **decouples high-level intent understanding from pixel-level coordinate regression** via two complementary tools.
- **High2Low instruction tool:** Rewrites coarse instructions into precise descriptions carrying widget type, anchor and region, removing intent ambiguity at the source.
- **Zoom-in resolution tool:** Partitions the screenshot into a patch grid, reasons about which patch holds the target, then zooms in to regress relative coordinates and the action (click, scroll, text input; 5 types), avoiding the drift of full-image regression.
- **Training & results:** Trained the High2Low model from Qwen2.5-VL with cold-start and GRPO, then distilled its instruction-specification capability into the grounding model; against the UI-R1-3B baseline, **ScreenSpot average rose 83.3 to 90.6 and ScreenSpot-Pro average 17.8 to 37.3, state-of-the-art**, with the largest gains on small icon/widget targets.

Publications & Research

Cognitive-structured Multimodal Agent (CMA) **First author** AAAI’27 under review

Episodic visual memory and retrieval-based context architecture for long-horizon multimodal agents (see Tencent internship, item 1). Paper Project Page Code

Modality-Balanced Decoupling Alignment for Video-Text Retrieval **First author** AAAI'27

- Proposed MBDA to close the semantic gap from **asymmetric modality information capacity and strongly coupled video representations**: modality proximity alignment pulls video embeddings toward text, while orthogonal decoupling separates aligned from modality-specific information; verified via low-rank decomposition and DFT analysis.
- Orthogonality confirmed by a drop in cosine similarity between aligned and modality-specific components from **0.052 to 0.0042**; ablations lift the MSR-VTT baseline from 42.1% to 49.2% and 47.8% for the two modules and 52.4% combined, giving **state-of-the-art retrieval on MSR-VTT / DiDeMo / MSVD / ActivityNet (52.4% / 53.1% / 54.0% / 49.6%)**.

H2L-G1: Advancing GUI Grounding with High-to-Low Tools **First author**

Tool-augmented GUI grounding; SOTA on ScreenSpot / ScreenSpot-Pro (see Baidu internship).

Semantic-Guided Generative Image Augmentation with Diffusion Models **AAAI 2024**

- Proposed SGID, a semantic-guided diffusion augmentation that builds prompts from image labels and captions to raise diversity while preserving semantics in Stable Diffusion image-to-image generation.
- Across 7 datasets and ResNet-50/ViT/CLIP-ViT backbones, improved over the strongest baseline by 1.72%, 0.33% and 0.14%, and by **10.39% when training ResNet-50 from scratch**; also studied RandAugment integration and beam-search / nucleus-sampling / CLIP selection.

Technical Skills

- **Post-training:** large-scale SFT, GRPO / DAPO reinforcement learning, cold-start + RL, capability distillation; reward design and downstream-task alignment (agent retrieval policies, GUI grounding)
- **Multimodal understanding & agents:** data construction, training, evaluation and deployment of VLMs (Qwen3-VL / Qwen2.5-VL); multimodal agent tool-use and memory retrieval; also unified bases (WeGen Omni / BAGEL) and image-editing models (Qwen-Image / Qwen-Image-Edit)
- **Training & inference infrastructure: foundation-model training at the thousand-GPU scale**, multi-node distributed training (ms-swift), vLLM tensor parallelism and multi-endpoint batch inference, dual-GPU CFG parallelism
- **Code-driven generation:** generation with **source code as the intermediate representation** — CoT emits HTML/CSS/SVG/restricted LaTeX and a typesetting engine renders deterministically; grammar-constrained decoding keeps the source valid, and rendering doubles as validation — failures are fed back with their errors for a rewrite
- **Data engineering:** headless Playwright rendering for synthesis, DOM-level bbox extraction (OCR-free annotation), seed- and style_spec-replayable samples, image-text pipelines at the ten-million scale, automated OCR/VLM quality gates with slice-level attribution
- **Languages & frameworks:** Python / PyTorch, C / C++, HTML / CSS / SVG, LaTeX, FastAPI, Gradio, OpenAI-compatible APIs

Projects & Contests

CUMCM National Second Prize: FAST Active Reflector Optimization *Team Lead* 2021.09

MCM Honorable Mention: Optimal Power Allocation in Cycling Races *2022.03*

National College Mathematics Contest (Non-math) Second Prize *2021.12*

Full-stack Encrypted Banking with Secure Transmission, Storage, and Authentication

Individual 2022.10 – 2022.11

Independently built a full-stack Java Web payment system securing transmission, storage and identity authentication via SET, symmetric/asymmetric encryption, digital envelopes and dual signatures.

Miscellaneous

- Scholarships: Guanghua (3/556), Tencent (8/556), First-class People's (top 3%), Honors School
- Open source: Top-AI-Conferences-Paper-with-Code on GitHub (2.6k stars)
- Student leadership: Chair of the Innovation Center, HIT Honors School; class monitor